ET AI HACKATHON 2026 • PROBLEM STATEMENT 9

ArthSaathi

India's AI-Powered Personal Finance Mentor

"A CA, a financial advisor, and a tax consultant — free, in 60 seconds, for every Indian with a smartphone."

14.2 Cr+	95%	₹25,000+	1 : 177
Demat accounts	Have no plan	Advisor fee/yr	Advisor ratio

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1. Project Overview

ArthSaathi is a multi-agent AI system that gives every Indian retail investor access to the same quality of financial advice previously available only to the wealthy. It is not a chatbot. It is six working product modules backed by real financial mathematics, Indian tax law, and behavioural science — all in one place.

1.1 The Problem ArthSaathi Solves

India has a severe financial advice gap that no existing product has bridged:

- 14.2 crore demat accounts exist in India, but most investors navigate without any professional guidance
- There is only 1 registered financial advisor for every 177 investors (SEBI 2024 data)
- A Certified Financial Planner charges ₹25,000 to ₹50,000 per year, making professional advice accessible only to High Net-worth Individuals
- Crores of retail investors make buy and sell decisions based on WhatsApp group tips from anonymous sources
- Most people do not know their true investment returns, how much they have paid in hidden commissions, or whether they are on track for retirement
- Behavioural biases like loss aversion and recency bias destroy wealth silently, and no tool addresses them at the interface level

1.2 The Solution at a Glance

ID	Module	What it does in one line
Feature 1	MF Portfolio X-Ray	Upload CAMS statement → true XIRR, overlap, hidden fees, rebalancing plan in 10 seconds
Feature 2	Tax Wizard	Upload Form 16 → old vs new regime comparison + every deduction you are missing
Feature 3	FIRE Path Planner	Enter life goals → month-by-month roadmap to financial independence with exact SIP amounts
Feature 4	Money Health Score	5-minute quiz → score across 6 financial dimensions with navigation to fixes
Feature 5	Life Event Advisor	Enter a life event → ranked, personalised action plan in exact rupee amounts
Feature 6	Couple's Money Planner	Both partners enter data → joint optimisation of HRA, NPS, SIP, and insurance
Unique 1	Cost of Bad Advice Detector	CAMS scan → exact rupees your agent earned from your portfolio

ID	Module	What it does in one line
Unique 2	Behavioral Bias Fingerprint	Quiz → your dominant bias → every output adapts its framing to your psychology
Unique 3	WhatsApp Tip Analyzer	Paste any tip → A-to-F grade with NSE volume spike check and red-flag analysis
Unique 4	Procrastination Cost Clock	Live real-time counter showing wealth lost every second you delay starting your SIP
Unique 5	The Mirror	Questionnaire answers vs actual CAMS behaviour → side-by-side reality check
Unique 6	Salary-to-Wealth Translator	Any rupee amount → its 30-year retirement corpus equivalent at 12% CAGR

2. Core Product Features

These six modules form the foundation of ArthSaathi. Each one is a fully working product that solves a specific and measurable financial problem for Indian retail investors.

F1 Feature 1 — MF Portfolio X-Ray

> *The single most powerful demo in the entire product. Most investors have no idea what their real annual return is. This feature tells them the truth in 10 seconds.*

What problem does this solve?

When an investor looks at their mutual fund app, they see “absolute return” — say 42% over 4 years. This sounds impressive. But 42% absolute over 4 years is only 9.1% per year. And if they made SIPs of different amounts at different times, even that number is wrong. The real return — called XIRR — could be much lower. Investors are being misled by the very metric displayed to them.

How does a user use this feature?

1. User logs in to CAMS website (camsonline.com) and downloads their Consolidated Account Statement as a PDF
2. User uploads this PDF to ArthSaathi

3. System parses every transaction — every SIP, switch, redemption, and dividend reinvestment — across all fund houses
4. System delivers a complete portfolio analysis in under 10 seconds

What the system shows

- True XIRR (annualised return): Calculated using the Newton-Raphson method via scipy.optimize. Each SIP payment is a negative cash flow at its exact date. Current portfolio value is a positive cash flow at today's date. The XIRR is the single interest rate that makes the net present value of all these cash flows equal to zero. This is the only accurate measure of investment performance across irregular contributions.
- Comparison table: Absolute return vs XIRR vs category benchmark index return. Example: "Your apparent return is 38%. Your true annualised return is 8.4%. The Nifty 50 index returned 12.1% over the same period."
- Overlap analysis: The system downloads the actual portfolio holdings of each fund from AMFI's free API. It then calculates pairwise Jaccard similarity between fund holdings weighted by portfolio allocation. If SBI Bluechip, Axis Bluechip, and Mirae Large Cap together hold HDFC Bank, Infosys, and Reliance as their top positions, the overlap is massive. The system presents this as: "Your portfolio has 73% overlap across 3 large-cap funds. You are paying for three funds but effectively owning one."
- Expense ratio drag: For each fund, the system calculates the total trail commission or expense ratio paid to date, and projects the cumulative cost over the next 20 years. A 1.5% expense ratio on a ₹10 lakh corpus compounding at 12% costs ₹13.2 lakhs more than a 0.1% direct index fund over 20 years.
- Benchmark comparison: Each fund is matched to its appropriate benchmark — large-cap funds vs Nifty 50, mid-cap vs Nifty Midcap 150, flexi-cap vs Nifty 500. Rolling 1-year, 3-year, and 5-year alpha and beta are shown.
- AI rebalancing plan: The Claude API receives the full analysis as structured JSON and generates a plain-English rebalancing plan: which funds to merge, which categories are underrepresented, suggested target allocation, and estimated improvement in portfolio efficiency after the change.

Technical implementation

PDF parsing uses the pdfplumber Python library. The CAMS statement format is structured and consistent. Transactions are extracted into a Pandas DataFrame with columns: date, fund name, transaction type (purchase/redemption/switch), units, NAV, amount. XIRR is solved using scipy.optimize.brentq with a search window of -0.99 to 10. AMFI holdings data is fetched from the free AMFI API endpoint. Overlap is computed using NumPy vectorised operations.

VIDEO DEMO STEPS

Step 1: Open the ArthSaathi website and navigate to the MF Portfolio X-Ray module

Step 2: Upload the pre-prepared sample CAMS PDF (included in the repo so judges can test instantly)

Step 3: Wait 8-10 seconds while the parsing and calculation engine runs — show the loading state

Step 4: Show the XIRR dashboard: headline number, comparison against apparent return and benchmark

Step 5: Scroll to the overlap analysis — highlight the 73% overlap across 3 large-cap funds

Step 6: Show the expense ratio drag chart: ₹47,000 paid in the last 5 years, ₹1.8 lakhs projected cost

Step 7: Click the 'Get rebalancing plan' button and show the AI-generated recommendation appearing

Step 8: Narrate: "This took 9 seconds. A professional portfolio review from a fee-only advisor costs ₹10,000 and takes a week."

F2 Feature 2 — Tax Wizard

The average salaried Indian leaves ₹20,000 to ₹40,000 in unclaimed tax deductions on the table every year. The Tax Wizard finds every rupee.

What problem does this solve?

The Indian income tax system has dozens of deduction sections, each with its own rules, limits, and conditions. Most salaried employees do not know all the deductions available to them. Their employer deducts TDS based on declarations made at the start of the year — often incomplete. The result is overpayment of tax by crores of salaried Indians every year.

How does a user use this feature?

5. User uploads their Form 16 PDF (provided by every employer at year-end) OR manually enters their salary structure
6. System parses Part A (TDS summary) and Part B (detailed income and deduction breakdown)
7. System asks 5 quick follow-up questions: Do you pay rent? Do you have a home loan? Do you pay health insurance premiums for parents? Are you contributing to NPS? What is your city?
8. System delivers complete tax analysis in under 15 seconds

What the system computes

- Old regime total tax: Gross salary minus all eligible deductions. Section 80C up to ₹1.5 lakhs (ELSS, PPF, EPF employee contribution, home loan principal, NSC, life insurance, children's school fees). Section 80D up to ₹25,000 for self and family health insurance plus ₹50,000 for senior citizen parents. Section 80CCD(1B) additional NPS contribution of ₹50,000 — this is completely separate from and in addition to the 80C limit. Section 24(b) home loan interest deduction up to ₹2 lakhs. HRA exemption calculated as the minimum of: actual HRA received from employer, 50% of basic salary for metro cities (40% for non-metro), and actual rent paid minus 10% of basic salary. Standard deduction of ₹50,000. Professional tax actually paid.
- New regime total tax: Only the revised standard deduction of ₹75,000 is allowed. Flat slabs: nil up to ₹3 lakhs, 5% from ₹3-7 lakhs, 10% from ₹7-10 lakhs, 15% from ₹10-12 lakhs, 20% from ₹12-15 lakhs, 30% above ₹15 lakhs. Section 87A rebate of up to ₹25,000 for income up to ₹12 lakhs.
- Side-by-side comparison: Tax payable under both regimes, difference highlighted, clear recommendation with explanation of why one regime is better for this specific income and deduction profile.
- Missing deduction detector: AI analyses the declared deductions against the user's profile. Common findings: 80C not fully used (user has capacity but is not investing), 80D for parents' health insurance not claimed (80,000 rupees of premium unclaimed = ₹24,000 tax saving at 30% bracket), NPS 80CCD(1B) not utilised (immediate ₹15,000 saving for a 30% bracket taxpayer), HRA calculated incorrectly by employer.
- Investment recommendations ranked by: (1) risk tolerance, (2) liquidity need, (3) expected return. ELSS mutual funds for those comfortable with 3-year lock-in and equity risk. PPF for those wanting safety and longer horizon. NPS for those focused on retirement. Fixed deposits or NSC for completely risk-averse.

VIDEO DEMO STEPS
Step 1: Navigate to the Tax Wizard module
Step 2: Select 'Enter manually' for a faster demo and enter sample salary data: ₹15 lakhs CTC, ₹40,000 HRA, pays rent of ₹20,000 in Bangalore
Step 3: Answer the 5 follow-up questions quickly: yes to rent, no home loan, parents have health insurance, no NPS contribution yet
Step 4: Show the old vs new regime comparison table appearing — old regime saves ₹34,000 in this scenario
Step 5: Show the missing deductions section: "NPS 80CCD(1B) not used — invest ₹50,000 in NPS to save additional ₹15,000 in tax"
Step 6: Show the ranked investment suggestion: ELSS SIP of ₹12,500/month fills the remaining 80C gap while giving equity returns

Step 7: Narrate: “This user was overpaying ₹49,000 in tax. That’s 4 months of SIP investment found for free.”

F3 Feature 3 — FIRE Path Planner

FIRE means Financial Independence, Retire Early. This planner answers the question every working Indian eventually asks: “When can I actually stop working?” It gives a specific date and a specific monthly SIP.

What problem does this solve?

Most people have vague financial goals like “retire comfortably” or “save for my child’s education.” Without numbers and a timeline, these goals produce no action. This planner converts vague goals into a precise month-by-month roadmap. It shows the user not just what to do but what happens if they do or do not do it.

Input collection

- Basic profile: Age, city, monthly post-tax income, monthly expenses broken into rent, food, transport, lifestyle, and EMIs
- Existing assets: EPF balance, PPF balance, mutual fund portfolio value, fixed deposits, stocks, gold, real estate equity
- Life goals: For each goal — name, target amount in today’s rupees, target year. Examples: child’s engineering degree 2040 ₹30 lakhs, home down payment 2027 ₹25 lakhs, retirement 2052 desired monthly income ₹75,000
- Risk profile: Conservative (FD-like returns), Moderate (balanced equity-debt), Aggressive (predominantly equity)

Calculations performed

- Future cost of each goal: Today’s goal amount inflated at the appropriate rate. Education inflation is assumed at 8% per year in India. General goals at 6%. Retirement corpus is calculated using the perpetuity formula: desired monthly income multiplied by 12 divided by the safe withdrawal rate of 4%. This gives the total corpus needed.
- SIP required per goal: Working backwards from the future cost, accounting for existing assets already accumulated towards that goal, and using the appropriate expected return rate for the risk profile and time horizon. Equity at 12% CAGR for horizons above 7 years, debt at 7%, blended for intermediate horizons.

- Emergency fund gap: Target is 6 months of total monthly expenses. Gap between current liquid savings and target is shown with a recommendation to invest in liquid mutual funds.
- Insurance gap: Human Life Value calculation — (annual take-home income) multiplied by (remaining working years) is the baseline. Existing life insurance cover is subtracted to find the gap. Health insurance gap is shown separately.
- Total monthly SIP needed: Sum across all goals. Shown alongside current monthly savings capacity (income minus expenses minus existing SIPs). If there is a shortfall, the system shows which goals to prioritise and which to defer.
- FIRE date: The age at which the total portfolio corpus crosses the threshold where a 4% annual withdrawal sustains the desired lifestyle indefinitely. Shown as a specific year and age.
- Interactive timeline chart: Portfolio value plotted against time, with goal milestones marked as vertical lines. User can drag SIP amount to see FIRE date change in real time.

VIDEO DEMO STEPS
Step 1: Navigate to FIRE Planner and enter: Age 32, income ₹1.5 lakhs/month, expenses ₹70,000/month, existing MF portfolio ₹8 lakhs
Step 2: Add three goals: child education in 2040 (₹30 lakhs today), home down payment in 2028 (₹25 lakhs), retirement at 52 (₹75,000/month desired)
Step 3: Select Moderate risk profile and click ‘Build my roadmap’
Step 4: Show the result: Total SIP needed ₹38,500/month. Breakdown per goal shown in a table.
Step 5: Show the FIRE date: age 52 if the user starts today. Scroll to the interactive timeline chart.
Step 6: Drag the SIP slider from ₹38,500 down to ₹25,000 and show the FIRE date shifting from 52 to 58 in real time
Step 7: Show the insurance gap: currently underinsured by ₹78 lakhs of life cover
Step 8: Narrate: “In 90 seconds, this person now has a complete financial blueprint that a CA would charge ₹15,000 to prepare.”

F4 Feature 4 — Money Health Score

> *The Money Health Score is the entry point of ArthSaathi. Every new user completes this in 5 minutes and gets a personalised score. The score tells them exactly where they are financially healthy and where they are vulnerable.*

The 6 dimensions

Dimension	How the score is calculated
Emergency preparedness	Target: 6 months of expenses in liquid account. Score = (months saved ÷ 6) × 100, capped at 100. Partial credit for amounts held in liquid funds.
Insurance coverage	Life score: current cover ÷ Human Life Value target. Health score: current cover ÷ 10x annual healthcare expenses. Combined, weighted 60% life, 40% health.
Investment diversification	Single fund in one category: 20 points. Multi-fund single category: 40. Multiple categories (equity + debt): 60. All asset classes including gold or international: 80. Globally diversified with index funds: 100.
Debt health	EMI-to-income ratio below 20%: 100 points. 20-30%: 75. 30-40%: 50. 40-50%: 25. Above 50% or credit card revolving balance: 0.
Tax efficiency	80C fully utilised: 30 points. Correct regime chosen: 30. NPS 80CCD contribution active: 20. HRA correctly calculated: 20. Maximum 100.
Retirement readiness	Current corpus as a % of total corpus required at retirement. Computed from FIRE formula. Being on track or ahead scores 100. Zero corpus at age 40 scores below 20.

How the score drives action

Each dimension card is interactive. Tapping on a weak dimension takes the user directly into the relevant module. Weak on insurance: goes to FIRE Planner insurance section. Weak on tax efficiency: goes to Tax Wizard. Weak on investment diversification: goes to MF Portfolio X-Ray. The score is not just a number — it is a navigation system.

VIDEO DEMO STEPS
Step 1: Navigate to Money Health Score and select ‘Take the assessment’
Step 2: Answer the 12 questions quickly, demonstrating that a typical salaried employee at age 32 would answer these in under 5 minutes
Step 3: Show the radar chart appearing with the 6 scores: Emergency 45, Insurance 22, Diversification 60, Debt 80, Tax 40, Retirement 35. Overall score: 47/100.
Step 4: Highlight the Insurance dimension in red at 22/100. Explain: “This person is underinsured by ₹75 lakhs — the single biggest financial risk in their life.”
Step 5: Click on the Insurance card and show it navigating directly into the insurance calculation inside the FIRE Planner
Step 6: Go back and click on Tax Efficiency at 40/100 — navigate into Tax Wizard
Step 7: Narrate: “The score is not a vanity metric. Every low score is a door into the fix.”

F5 Feature 5 — Life Event Financial Advisor

Life events are the moments when the most money is won or lost. A bonus invested wrong. A home loan structured poorly. An inheritance taxed unnecessarily. ArthSaathi catches these moments and gives the right advice at the right time.

Supported life events and what happens for each

Life event	What ArthSaathi delivers
Annual bonus received	User enters bonus amount, current loans, emergency fund status. System shows optimal split: clear high-interest debt first (credit card > personal loan), top up emergency fund if below 3 months, invest surplus per FIRE plan allocation. Exact rupee amounts for each.
Inheritance or windfall	User enters amount and source. System covers: gift tax rules (transfers from specified relatives are tax-free), LTCG on any assets inherited, phased deployment via STP for large equity amounts to avoid timing risk, and estate planning considerations.
Getting married	Merge income and expense profiles. Optimise HRA claims across both salaries. Review and consolidate overlapping insurance policies. Update nominees on all investment accounts. Suggest joint financial goals.
New baby	Calculate education corpus target (engineering degree in 2043 will cost ₹60-80 lakhs). Recommend starting ELSS SIP immediately for 18-year compounding runway. Add child to floater health insurance. Increase term life cover to account for the new dependent.
Buying a home	Optimal loan structure given income and tenure. Section 24(b) interest deduction calculation. Whether to prepay the loan or invest surplus — the break-even analysis compares home loan rate vs expected investment return.
Job change or loss	Emergency fund runway. EPF transfer checklist and process. Health insurance bridge recommendation for the gap period. ESOP exercise decision framework. Tax implications of dual employment in the same financial year.

VIDEO DEMO STEPS

Step 1: Navigate to Life Event Advisor and select ‘Annual bonus received’

Step 2: Enter: Bonus ₹5 lakhs, existing personal loan ₹4.5 lakhs at 14% interest, emergency fund covers 1.5 months

Step 3: Show the ranked action plan: ₹1.5 lakhs clears the personal loan (“risk-free 14% return”), ₹1.8 lakhs tops up emergency fund to 3 months, ₹1.7 lakhs invested in ELSS (fills 80C gap and aligns with FIRE plan)

Step 4: Show the tax saving: the ₹1.7 lakh ELSS investment saves ₹51,000 in tax this year at 30% bracket

Step 5: Quickly demo the 'New baby' scenario: enter child's birth year 2024, target IIT education

Step 6: Show the output: start ₹7,200/month ELSS SIP today. At retirement that corpus becomes ₹61 lakhs.

Step 7: Narrate: "Every major financial decision in a person's life is covered. This is the advisor who shows up at the right moment."

F6 Feature 6 — Couple's Money Planner

India's first AI tool for joint financial planning. When two incomes are in play, there are specific optimisations available that no individual planning tool can find. ArthSaathi finds every single one.

Joint optimisations the system finds

- HRA optimisation: If both partners pay rent (even in the same house), both can legally claim HRA. The system calculates the optimal rent declaration split based on which partner gets a higher per-rupee HRA benefit after comparing their basic salary, HRA component, and tax bracket. The total household tax saving can be ₹30,000 to ₹75,000 per year.
- NPS employer matching: If one partner's employer offers NPS matching, that partner's NPS contribution must be maximised first before any other voluntary investment because employer NPS contribution is additional income plus tax-free return. The system prioritises this before all other recommendations.
- SIP split for LTCG efficiency: Long-term capital gains on equity mutual funds above ₹1 lakh per year are taxed at 10%. Each individual has their own ₹1 lakh exemption. If both partners hold mutual funds separately, the household effectively has ₹2 lakhs of LTCG exemption per year. The system optimises which funds are held in whose name based on expected redemption year and amount.
- Insurance joint review: Compares two individual health insurance policies against a single family floater plan. The family floater is usually cheaper by 20-30% for the same total cover. The system calculates and recommends the lower-cost option.
- Combined net worth dashboard: Single unified view of total household assets (mutual funds, EPF, PPF, FDs, stocks, real estate equity, gold), liabilities (home loan, car loan, personal loan), insurance coverage for both, and progress towards all shared goals.

VIDEO DEMO STEPS

Step 1: Navigate to Couple's Money Planner and enter both partners' details

Step 2: Partner A: ₹18 lakhs annual, pays ₹25,000 rent in Mumbai, employer offers NPS matching

Step 3: Partner B: ₹12 lakhs annual, same address, also claims HRA

Step 4: Show the HRA optimisation result: “Split rent declaration as 60% Partner A, 40% Partner B. Combined HRA saving: ₹58,000 per year.”

Step 5: Show the NPS matching alert for Partner A: “Your employer matches up to ₹60,000 in NPS. This is free money. Maximise this before any other investment.”

Step 6: Show the combined net worth dashboard with progress bars for each shared goal

Step 7: Narrate: “Two individual financial plans miss these optimisations. Only a joint plan finds them.”

3. Unique Features Found Nowhere Else in India

These six features are not available in any Indian fintech product today — not in Zerodha, Groww, ET Money, Kuvera, INDMoney, or any other platform. Each one is built on a genuine insight about Indian investor behaviour that incumbent products cannot or will not address.

U1 Unique Feature 1 — Cost of Bad Advice Detector

Why no other product has this: Every major fintech platform in India earns commission from Regular plan mutual funds. Showing users the cost of this commission would destroy their own revenue. ArthSaathi has no such conflict.

The problem

In India, mutual funds are sold in two variants. Direct plans have no distributor and lower expense ratios — typically 0.1% to 0.5% per year. Regular plans pay a trail commission to the distributor or platform and have expense ratios of 0.9% to 2.0% per year. The difference is not disclosed to the investor in any obvious way. Over 20 years, this silent gap compounds into lakhs of rupees extracted from the investor’s wealth.

How it works

- After the CAMS X-Ray parses the statement, the system identifies every fund held in Regular plan format
- For each Regular plan holding, it fetches the equivalent Direct plan expense ratio from AMFI data

- The expense ratio difference is multiplied by the investment amount and compounded over the holding period to calculate the exact rupee cost paid to date
- The same calculation is projected forward 20 years to show total lifetime cost
- The Direct plan equivalent for every fund is shown side-by-side
- A “switch to Direct” guide is generated, noting that switching triggers LTCG/STCG, so the optimal time and method is explained

> *The demo line: “Your mutual fund distributor has earned ₹47,000 from your portfolio in the last 5 years. You could have had that money working for you instead.”*

VIDEO DEMO STEPS
Step 1: After completing the MF Portfolio X-Ray demo, scroll to the ‘Cost of Bad Advice’ section
Step 2: Show the table of all Regular plan holdings with their expense ratios highlighted
Step 3: Show the side-by-side: Regular plan expense ratio 1.65% vs Direct plan equivalent 0.42%
Step 4: Show the rupee cost to date: ₹47,300 paid in distributor commissions over 5 years
Step 5: Show the 20-year projection: staying in Regular plans will cost ₹3.8 lakhs in additional fees
Step 6: Show the switch guide: “Wait until January 2025 to switch these two funds to avoid short-term capital gains tax this financial year”
Step 7: Narrate: “No other Indian platform shows this because they are the ones earning the commission.”

U2 Unique Feature 2 — Behavioral Bias Fingerprint

> *Why no other product has this: Applying behavioural finance at the interface layer — where every output silently adapts its framing based on the user’s detected psychology — has never been done in Indian retail finance.*

The five biases detected

Bias	How it shows up	How ArthSaathi adapts
Loss aversion	Holds losing funds for years rather than cutting losses. Refuses to redeem anything below their buy price. Classic manifestation: still	Rebalancing shown as ‘locking in gains on winners’. Insurance gaps shown as losses avoided, not purchases made. Cost of inaction always shown alongside the action.

Bias	How it shows up	How ArthSaathi adapts
	holding an NFO from 2018 that is down 40%.	
Recency bias	Invests in whatever was the top performer last year. Consistently chases momentum without understanding mean reversion.	10-year returns shown first, 1-year last. Warning label on any fund that was top-ranked last year. Historical return of rank-1 fund 3 years later always shown.
Overconfidence	Trades frequently, switches funds often, holds concentrated single-stock bets.	Lump sum replaced with STP recommendations. Data on individual stock-picker performance vs index shown. Switching history cost calculated.
Herd mentality	Buys NFOs and thematic funds based on social proof. Invests in whatever friends are talking about.	NFOs explicitly flagged. Thematic funds shown with their 5-year underperformance vs broad index. Diversification score shown prominently.
Anchoring	Refuses to sell or rebalance until a fund 'comes back' to its original purchase price. NAV anchoring is the most common investor mistake.	Opportunity cost of waiting shown: 'If this fund returns to your buy price in 3 years, you will still have earned only 2.4% CAGR. The Nifty would have returned 12%.'

VIDEO DEMO STEPS

Step 1: Navigate to the onboarding quiz and complete the 10-question behavioral assessment

Step 2: Answer the questions in a way that demonstrates loss aversion (hold losers, hate selling at a loss)

Step 3: Show the Bias Fingerprint radar chart: high on Loss Aversion, moderate on Recency Bias

Step 4: Show the MF X-Ray output: notice the rebalancing recommendation is now framed as 'Lock in your winners' not 'Cut your losers'

Step 5: Show a fund recommendation card: returns displayed as 10-year CAGR prominently, 1-year return in small text below

Step 6: Reset the quiz, answer as an overconfident trader

Step 7: Show how the same rebalancing recommendation now emphasises data on active trading costs and index fund outperformance

Step 8: Narrate: "Same analysis, completely different psychological framing. This is why behavioural finance actually changes behaviour."

U3 Unique Feature 3 — WhatsApp Tip Analyzer

> *Why no other product has this: Defending retail investors against market manipulation is not profitable. The WhatsApp Tip Analyzer does what no Indian fintech has been willing to build.*

The problem

The SEBI enforcement division consistently finds that pump-and-dump schemes in Indian small and micro-cap stocks are coordinated through WhatsApp and Telegram. The typical pattern: an operator accumulates a stock, then floods investor groups with a tip containing a specific price target. Retail investors buy, price spikes, operator sells at the top. Retail investors are left holding a depreciating asset.

The 6-point analysis

- NSE volume spike detection: The system queries NSE's free bhavcopy data (daily price and volume file) for the stock mentioned in the tip. It calculates the 20-day average volume and compares it to the 5 days before the tip appears to have circulated. A 3x or greater volume spike in the 3-5 days before the tip is a strong pump-and-dump indicator.
- Language red-flag scoring: The AI analyses the tip text for high-risk phrases: 'operator entering', 'rumoured buyback', 'must buy', '100% return guaranteed', 'insider info', 'burn your shoes'. A weighted score is computed based on phrase category and count.
- F&O ban status: Stocks in the NSE securities-in-ban period cannot be traded in derivatives by institutions. A tip on a banned stock is suspicious because legitimate research does not target untradeable securities.
- Analyst coverage check: The system checks whether the stock has coverage from any SEBI-registered research analyst. A specific price target on a stock with zero analyst coverage is a red flag for manufactured hype.
- Promoter pledge check: If promoter shareholding pledge is above 50%, the stock is financially distressed. Tips on such stocks often come from promoters trying to create retail demand.
- Final grade A to F: A = credible with public data alignment. B = possible but verify. C = multiple warning signs. D = high probability of manipulation. F = classic pump-and-dump pattern. Accompanied by plain-English explanation of each finding.

VIDEO DEMO STEPS
Step 1: Navigate to WhatsApp Tip Analyzer
Step 2: Paste a sample tip (pre-prepared for demo): 'BUY ALERT: XYZ Microcap Ltd, CMP 47, Target 140 in 3 months. Strong operator activity. Promoter buying. Rumoured SEBI-approved expansion. Burn your shoes on this one.'
Step 3: Wait 3 seconds for analysis to complete
Step 4: Show Grade F result with the full breakdown: volume spike of 4.2x detected 4 days before tip, 3 red-flag phrases identified, zero analyst coverage, promoter pledge at 68%

Step 5: Paste a second sample tip about a legitimate large-cap stock with recent DRHP filing

Step 6: Show Grade B result: no volume anomaly, credible business news, but still note 'verify with official exchange filing'

Step 7: Narrate: "14 crore investors get tips like this every day. This is the only tool in India that tells them whether to act on it or ignore it."

U4 Unique Feature 4 — Procrastination Cost Clock

Why no other product has this: Making an abstract future cost visible and counting in real time is one of the most powerful behavioural nudges known to psychology. No fintech in India has applied it to investment delay.

The mechanism

After the FIRE Planner calculates the user's required monthly SIP, a live counter appears on the dashboard. The counter increments every second, showing the total future retirement wealth the user is forgoing by not starting the SIP today.

The mathematics

If the required SIP is ₹30,000 per month and the time horizon is 22 years at 12% CAGR, then the future value of the entire SIP stream is approximately ₹3.95 crore. Divided by the number of months (264), each month's SIP creates approximately ₹14,962 of future wealth. Divided by 30 days and then by 86,400 seconds, each second of delay costs approximately ₹5.77 in future wealth. The counter shows this accumulating.

The psychological effect

This mechanism is drawn from two well-established behavioural science findings. First, making an abstract future consequence concrete and real-time dramatically increases the salience of present decisions (Kahneman, Thinking Fast and Slow). Second, loss framing — showing what is being lost rather than what could be gained — is approximately twice as motivating for most people (Prospect Theory). The counter combines both: it makes future loss concrete and shows it accumulating in real time.

The resolution event

The moment the user sets up a SIP through ArthSaathi's partner link, the counter freezes and a badge appears: "You locked in ₹1,12,45,600 in retirement wealth today." The number is their total

projected retirement corpus from this SIP. The framing shifts from loss to gain at the moment of action.

VIDEO DEMO STEPS
Step 1: Complete the FIRE Planner with the required SIP of ₹38,500 per month
Step 2: Show the dashboard with the Procrastination Cost Clock visible in the top right
Step 3: Point to the counter ticking upward: “Every second this counter is running, this person is forgoing ₹6.80 in retirement wealth”
Step 4: Let it run for 10 seconds — counter moves from ₹0 to approximately ₹68
Step 5: Extrapolate for the audience: “In one day of delay this person loses ₹587. In one year, ₹2.14 lakhs. In the decade they have been meaning to start their SIP, ₹21 lakhs.”
Step 6: Click ‘Set up my SIP now’ — counter freezes. Badge appears showing total locked-in retirement wealth
Step 7: Narrate: “This is not a guilt trip. This is the truth made visible. The moment they act, the loss stops.”

U5 Unique Feature 5 — The Mirror: Stated vs Actual Behavior

> *Why no other product has this: Doing this requires both a questionnaire and access to actual transaction history. Only ArthSaathi has both. And only ArthSaathi is honest enough to show the contradiction.*

The problem

During risk profiling, investors reliably overstate their risk tolerance. They have been conditioned by financial advisors who ask: “are you a long-term investor?” to answer yes. Their actual transaction history from their CAMS statement tells a completely different story. Reconciling the two produces the most accurate risk profile available.

The Mirror output

What you said	What your transactions show
“I am a long-term investor with a 7-10 year horizon”	Average holding period across all funds: 11 months
“I do not panic in market downturns”	Redeemed ₹2.3 lakhs in March-April 2020 (COVID crash bottom)
“I prefer equity over debt”	63% of current portfolio is in debt and liquid funds

"I have an aggressive risk profile"	Actual revealed preference is Conservative-Moderate
Claimed: Comfortable with 30% temporary loss	Reality: Sold at 18% loss in 2020, never re-entered

After displaying the Mirror, the AI says: "Your actual risk tolerance, based on your behaviour rather than your stated preference, is Conservative-Moderate. Your optimal portfolio should have 50-60% equity and 40-50% debt. Here is a rebalancing plan that matches who you actually are."

VIDEO DEMO STEPS

Step 1: This feature flows naturally from the MF Portfolio X-Ray — begin immediately after showing the X-Ray

Step 2: Show the risk profiling questionnaire with the user having answered 'Aggressive' on all dimensions

Step 3: Scroll down to The Mirror section on the same page

Step 4: Show the side-by-side table: stated aggressive vs actual 11-month average holding period

Step 5: Highlight the March 2020 redemption row — explain: "The stock market bottomed on 24th March 2020. This user sold at the exact bottom. Their questionnaire says they don't panic."

Step 6: Show the revised risk profile and the new portfolio recommendation based on actual behaviour

Step 7: Show the difference in expected outcomes: the Conservative-Moderate portfolio has lower volatility and, for this specific investor, is more likely to be held through a crash

Step 8: Narrate: "The Mirror is not a judgment. It is the most honest financial assessment this person has ever received."

U6 Unique Feature 6 — Salary-to-Wealth Translator

> *Why no other product has this: Most fintech products want users to feel their money is too small to matter. This tool shows that every rupee, at the right return over a long horizon, matters enormously.*

The concept

Every financial decision has a cost denominated not in today's rupees but in the retirement corpus it creates or destroys. Most people make decisions in monthly units. The Salary-to-Wealth Translator converts any monthly or annual amount into its retirement wealth equivalent, making every financial trade-off visible in the only number that ultimately matters.

Three modes

- Salary increment mode: Enter the size of your next salary hike or a job offer's income increase. The system computes its retirement wealth equivalent at 12% CAGR over the remaining working years. Example: "A ₹1 lakh annual increment (if invested), for a 32-year-old retiring at 60, is worth ₹2.98 crore at retirement. That salary negotiation conversation is worth ₹3 crore."
- Expense trade-off mode: Enter any monthly spending category. The system shows its retirement wealth equivalent. Example: "Your ₹15,000/month dining and food delivery spend, if redirected to a SIP, would create ₹4.47 crore over 28 years at 12% CAGR. You are trading ₹4.47 crore for restaurant meals." This is not moralising — it is information. The user decides. But now they decide with full knowledge of the trade-off.
- Side project / income mode: Enter any additional income stream — freelance income, rental yield, dividend income. The system shows its lifetime wealth creation potential. Example: "Your ₹25,000/month freelance project, invested for 20 years, creates ₹2.5 crore. You are not running a side hustle. You are building a ₹2.5 crore business."

The Salary-to-Wealth Translator is also the most shareable feature in ArthSaathi. People screenshot the output and share it. "I just learned my gym membership costs me ₹47 lakhs in retirement wealth" is a tweet that writes itself. This is organic word-of-mouth growth built into the product.

VIDEO DEMO STEPS
Step 1: Navigate to the Salary-to-Wealth Translator (accessible as a widget on the FIRE Planner dashboard)
Step 2: Salary increment mode: Enter current salary ₹18 lakhs, expected hike to ₹21 lakhs, age 32
Step 3: Show the output: "This ₹3 lakh increment, if invested for 28 years, is worth ₹8.94 crore at retirement. Your next appraisal conversation is worth ₹9 crore."
Step 4: Switch to expense trade-off mode: Enter ₹18,000/month on dining out
Step 5: Show: "This spend, if redirected to a SIP, creates ₹5.37 crore. You are choosing ₹5.37 crore vs restaurant meals."
Step 6: Switch to side income mode: Enter ₹30,000/month freelance writing income
Step 7: Show: "This income stream, invested for 25 years, creates ₹3.97 crore. This is a ₹4 crore business."
Step 8: Narrate: "This feature makes every financial trade-off visible. Empowered decisions require visible consequences."

4. System Architecture

4.1 The Multi-Agent Design

ArthSaathi is not a single AI answering questions. It is a coordinated system of specialist agents, each with deep expertise in a specific financial domain. This architecture makes outputs more accurate, the system more auditable, and each agent replaceable or upgradeable independently.

Agent	Responsibilities
Orchestrator	The central coordinator. Receives user input, determines which specialist agents to invoke, sequences their execution, combines outputs, maintains conversation context, and logs every decision for auditability.
Portfolio agent	Owns all financial calculation tools: XIRR engine, overlap calculator, expense ratio analyser, benchmark comparator, and rebalancing logic. Queries AMFI free API and NSE historical bhavcopy data.
Tax agent	Holds the complete Indian income tax rule engine for FY 2024-25. Knows all sections from 80C to 87A, both regime slab structures, all exemption limits, and HRA calculation logic.
Planning agent	Financial modelling. Handles goal-based SIP projections, FIRE date calculation, asset allocation by time horizon, HLV computation, and emergency fund sizing. Uses RBI inflation targets and NSE historical CAGR data.
Insurance agent	Computes Human Life Value, identifies life and health coverage gaps, and generates gap amounts in crores. Knows IRDAI norms for recommended coverage levels.
Bias agent	Processes onboarding quiz results and CAMS transaction history. Generates the bias fingerprint and passes bias parameters to the orchestrator to modify framing of all outputs throughout the session.
Compliance agent	Runs on every output before it reaches the user. Ensures outputs are information and education, not regulated investment advice. Adds SEBI-mandated disclaimers. Blocks specific investment recommendations without completed risk profiling.
Tip analysis agent	Handles WhatsApp tip analysis. Queries NSE bhavcopy for volume data, runs the language red-flag model, checks F&O ban status, verifies analyst coverage, and computes the A-to-F grade.

4.2 Technology Stack

Layer	Choice
Frontend	React 18, TypeScript, Vite, Tailwind CSS, Recharts (financial charts), Zustand (state)
Backend	FastAPI (Python 3.11) — chosen for scipy, numpy, pdfplumber, pandas
AI layer	Claude claude-sonnet-4-20250514 via Anthropic API with structured JSON output
PDF parsing	pdfplumber (primary), camelot-py (fallback for complex tables)
Financial math	scipy.optimize for XIRR, numpy for vectorised calculations
Data sources	AMFI free NAV API, NSE bhavcopy CSV, RBI data, Income Tax Act FY 2024-25
Hosting	Vercel (frontend), Render or Railway (FastAPI backend)
Privacy	All uploaded documents processed in-memory only, never stored to disk

5. Business Impact Model

5.1 Value Created Per User Per Year

Value driver	Estimated benefit per user
Tax optimisation (Tax Wizard)	Average missed deductions ₹1.2 lakhs → tax saved ₹24,000 to ₹36,000
Regular → Direct plan switch	0.7% annual saving on ₹10 lakh portfolio → ₹7,000/year in lower fees
Better asset allocation (FIRE)	1% better CAGR on ₹25K/month SIP over 25 years → ₹45 lakhs more at retirement
Insurance gap closure	Prevents catastrophic family financial loss. Average gap: ₹75 lakhs to ₹1.5 crore
WhatsApp tip protection	Average loss from acting on one bad tip: ₹18,000 to ₹45,000
HRA couple optimisation	Typical dual-income couple saving: ₹30,000 to ₹75,000 per year in reduced tax

5.2 Market Size

Metric	Number
Total addressable market	14.2 crore demat account holders in India (SEBI 2024)

Metric	Number
ET serviceable market	3-4 crore ET Digital monthly active users
Revenue potential at 5% conversion	If 5% of ET users pay ₹3,000/year for ET Premium with ArthSaathi: ₹450-600 crore annually
Comparison benchmark	Same advice from a human CFP: ₹25,000 to ₹50,000 per year, serving HNIs only

6. Complete 3-Minute Video Demo Script

This is a suggested flow for the hackathon pitch video. All 12 features are referenced. Core features 1-3 and Unique features 3, 4, and 5 are shown live. Others are shown briefly or called out as the camera pans.

Opening (0:00 - 0:20)

Open on the landing page. Narrate: “India has 14.2 crore demat accounts and only 8 lakh financial advisors. That means 13.6 crore investors are completely on their own. ArthSaathi changes that. It is a full-stack AI financial mentor — 12 features, all working, all built on real Indian financial law and real data.”

Money Health Score (0:20 - 0:45)

Open the Money Health Score and quickly answer the quiz. Show the radar chart appearing. Overall score 47/100. Narrate: “The first thing every user sees. A complete financial health diagnosis in 5 minutes.” Pan over each of the 6 dimension scores.

MF Portfolio X-Ray (0:45 - 1:20)

Upload the sample CAMS PDF. Show the 9-second load. Narrate: “Real CAMS statement. Real math.” Show the XIRR comparison — apparent return 38%, true XIRR 8.4%, benchmark 12.1%. Show the 73% overlap finding. Show the Cost of Bad Advice: ₹47,300 in commissions. Narrate: “No other app in India shows this. They earn the commission.”

Tax Wizard (1:20 - 1:45)

Quickly enter Form 16 data. Show old vs new regime. Highlight the missing NPS deduction finding. Show ₹49,000 in annual tax saving identified. Narrate: “The average Indian leaves ₹30,000 to ₹50,000 in tax on the table every year. This finds it in 15 seconds.”

WhatsApp Tip Analyzer (1:45 - 2:05)

Paste the sample pump-and-dump tip. Wait 3 seconds. Show Grade F with the volume spike chart — 4.2x volume spike 4 days before tip. Show the red-flag phrases highlighted. Narrate: “The most unique thing we built. 14 crore investors have never had this protection before.”

The Mirror (2:05 - 2:20)

Show the stated vs actual behaviour table from the CAMS analysis. Highlight the March 2020 redemption at the market bottom. Narrate: “This person said they don’t panic. Their transactions say otherwise. The Mirror is the most honest financial analysis this person has ever received.”

Procrastination Cost Clock (2:20 - 2:30)

Show the live counter ticking upward on the FIRE Planner dashboard. Let it run for 5 seconds. Narrate: “Every second this user delays starting their SIP, they lose ₹6.80 in retirement wealth. The counter stops the moment they act.”

Salary-to-Wealth Translator (2:30 - 2:40)

Quickly enter a salary increment of ₹3 lakhs per year. Show the output: worth ₹8.94 crore at retirement. Show the dining expense: ₹18,000/month = ₹5.37 crore. Narrate: “Every financial trade-off, made visible.”

Closing (2:40 - 3:00)

Show the full feature dashboard with all 12 modules visible. Narrate: “Every feature you just saw uses real math, real Indian tax law, and a real CAMS statement. This is not a demo. This is production-ready software. Every other tool tells you what to do with your money. ArthSaathi is the first one that tells you what your money is already doing to you — without your knowledge.”

7. Hackathon Build Plan

Day	Build focus
Day 1	CAMS PDF parser, XIRR engine, Portfolio X-Ray display with overlap and benchmark comparison. Demoed end-to-end before midnight.
Day 2	Tax Wizard: Form 16 parser, both regime calculations, missing deduction AI. Cost of Bad Advice detector (low cost — uses already-parsed CAMS data).
Day 3	Money Health Score questionnaire and radar chart. WhatsApp Tip Analyzer (highest-impact unique feature per demo minute).
Day 4	FIRE Path Planner with interactive timeline chart. The Mirror (low engineering cost — uses CAMS + quiz data already available). Procrastination Cost Clock.
Day 5	Salary-to-Wealth Translator. FIRE Planner polish. Compliance agent. Prepare demo CAMS file, record pitch video, write README and architecture doc.
Roadmap (not built)	Couple's Planner and Life Event Advisor mentioned in README and architecture doc as designed and planned. Present the design clearly. Judges evaluate thinking, not just code.

The one sentence that ties everything together

"Every other tool tells you what to do with your money.
ArthSaathi is the first one that tells you
what your money is already doing to you — without your knowledge."

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